

Are Google Play Store App Reviews Being Written by AI?

Summary of Key Findings/Quick Takeaways

- Almost every single app had some percentage of AI generated content
- Entertainment & media apps had the highest AI content with 30.3%
- Overall, 16% of all app reviews are suspected to be AI generated

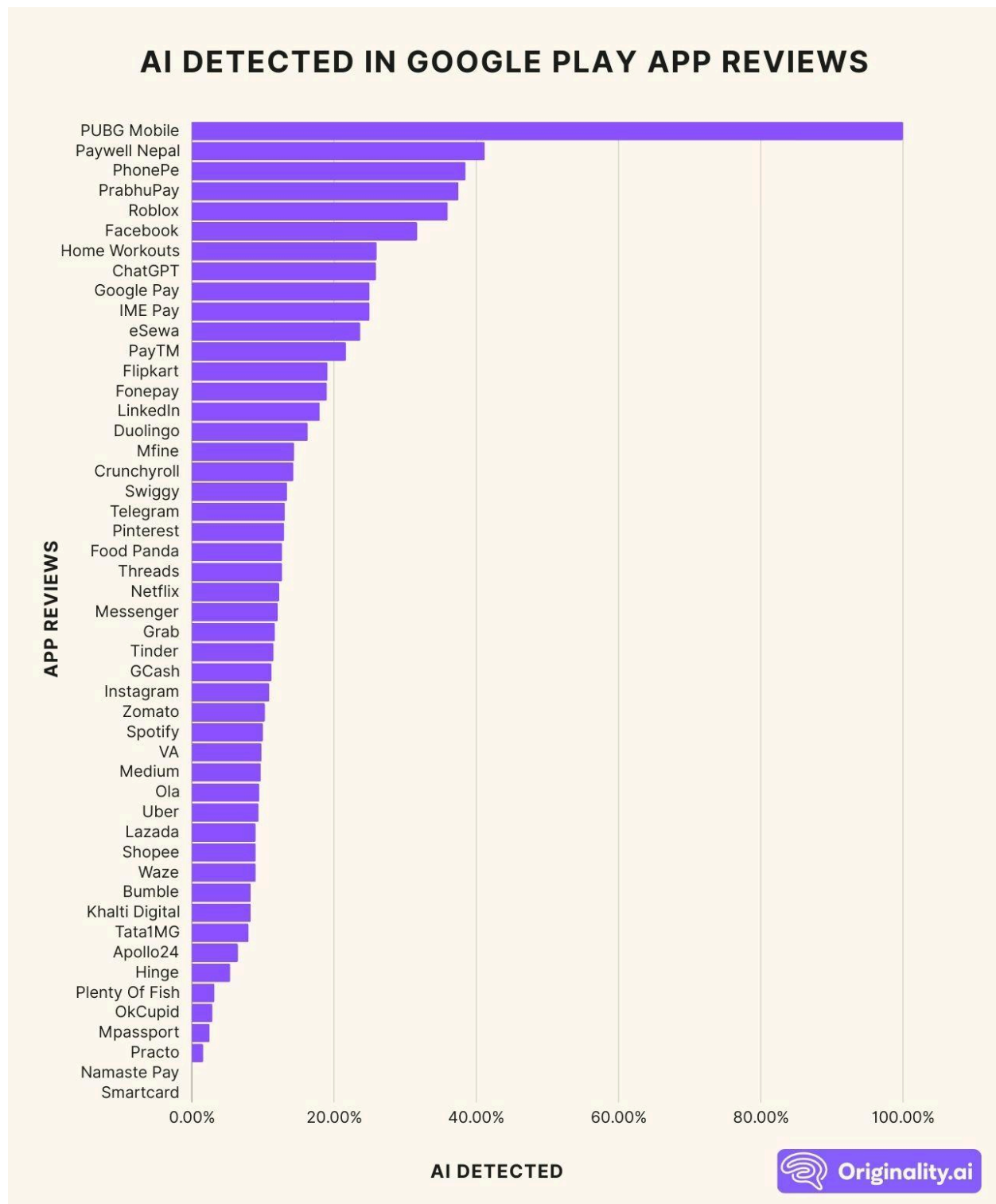
The emergence of artificial intelligence in recent years has been very popular and it is used by many people on a daily basis. AI can be used as a tool to boost productivity and increase efficiency by automating repetitive tasks. Things such as data entry and processing, web scraping, sales forecasting, and data analysis can be done with the help of AI. While AI offers a lot of benefits and increases our workflow, there are some areas where the work should be done by a real person. One of these areas is writing reviews for applications, as it involves writing about experiences, as well as pros and cons of the application. Reviews are important as they allow the user to share their experiences, as well as provide feedback to other users.

In this study we focused on analyzing AI generated content in Google Play Store app reviews. By analyzing multiple data sets and reviews, we aim to figure out which percentage of reviews from 2023-2024 are AI-generated on the Google Play Store.

Study Method and Data

The analysis consists of 50 applications and roughly 200 reviews for each application on the Google Play store. The datasets were found on Kaggle, and each dataset was filtered to include reviews from 2023-2024. In addition, the reviews were scraped to only include the ones that were greater than 50 words and containing no emojis. Each review was then scanned using the Originality AI API scanner and given a human score and an AI score. The findings are explained below.

Our Analysis (AI vs Human Score)

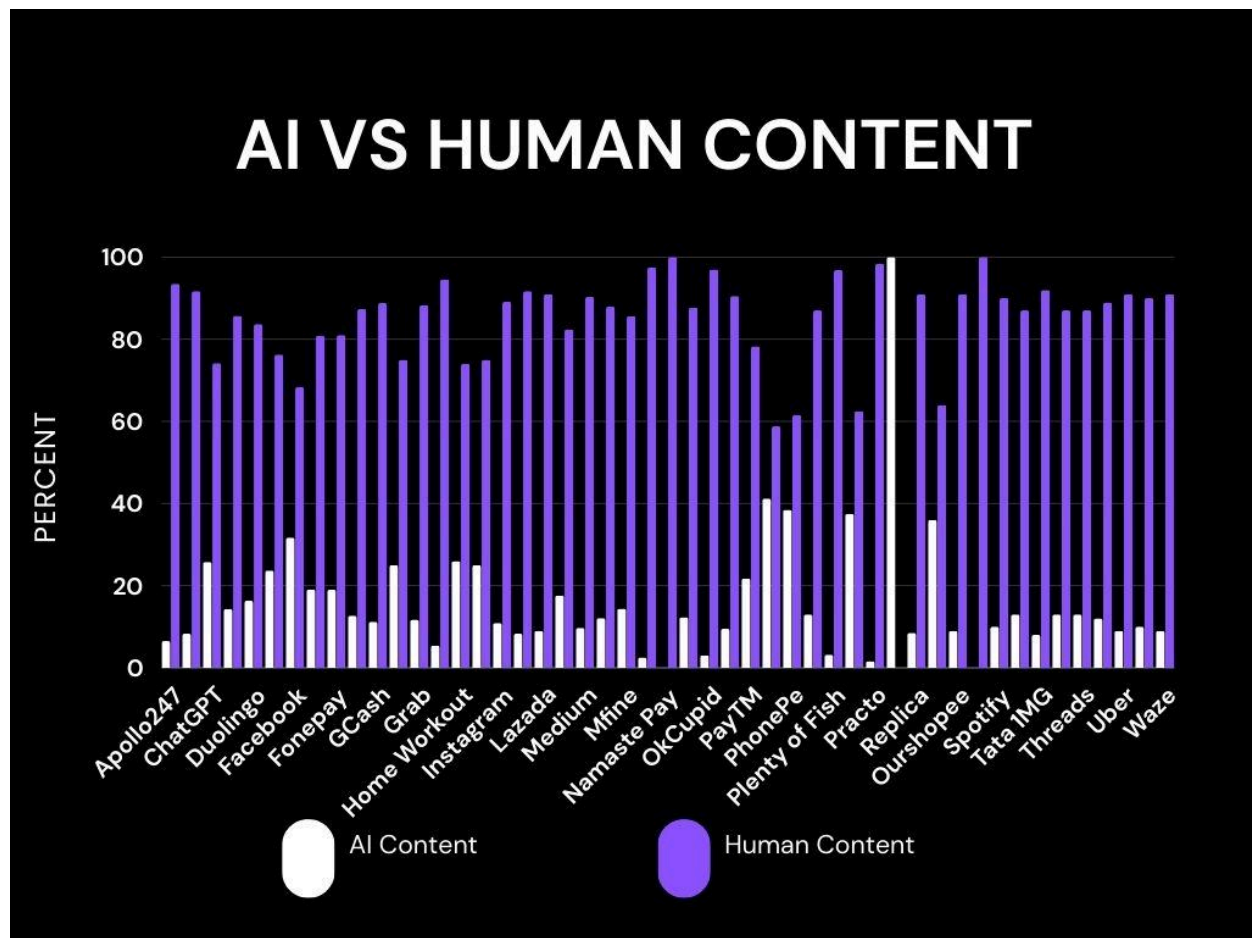


In each dataset, the reviews were individually scanned and given a score. Scores ≥ 50 were labelled as AI content while anything < 50 was labelled as human content. After all the reviews were scanned and labelled, the average AI score for the dataset was calculated (AI Content /

Total Reviews). These are the numbers that are depicted in the graph. When looking at the average of all the datasets, each dataset was found to have a 16.5 % AI score.

Let's add an short breakdown of the graph above... i.e. "In the graph above you can see PUBG mobile had 100% of it's reviews detected as AI generated, while the next 10 all had above 20% of their reviews AI generated." Any assumptions around this? Did you see any patterns in why these specific apps may have had AI generated reviews vs others that didn't have as much?

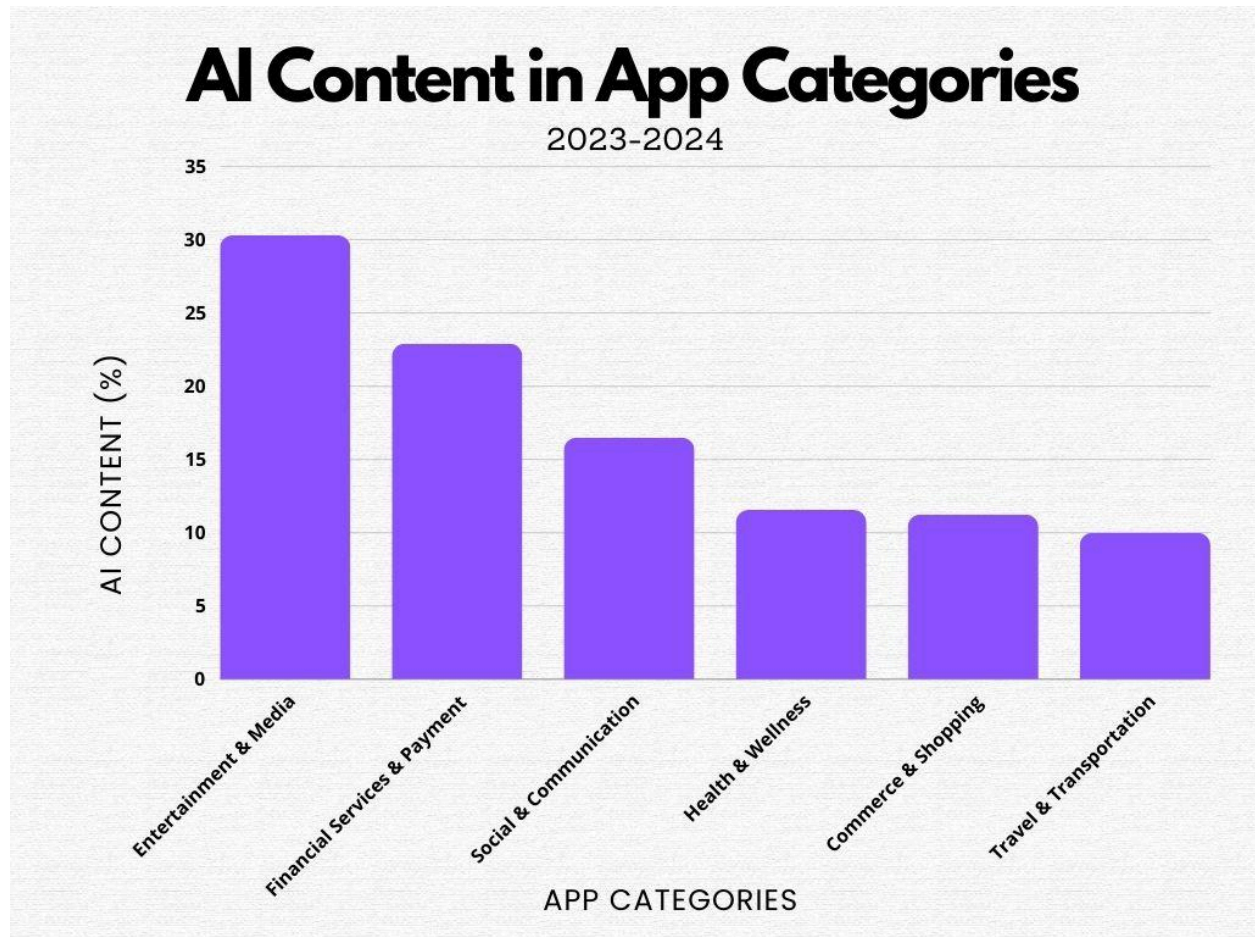
When analyzing the results of the graph, it is clear that the scanner does provide some false positives. Out of 176 reviews, it is highly unlikely that all of them are generated by AI. It is important to keep in mind that the scanner may not always be fully accurate. The apps that had more AI generated reviews were the ones that were games (entertainment & media) and financial payment apps. One possible reason for this could be that the population of users who left these reviews may be younger and more likely to use AI in their everyday lives. They could have used AI to write the reviews, rather than writing it themselves.



This figure shows both the AI and human content for each of the Google Play apps that were analyzed. Each app has both a human and AI score. It is evident that the majority of apps have some percentage of AI generated content.

App Categories vs AI Score

Different applications had different AI scores. To better understand the AI scores and their relevance, the correlation between AI scores and categories of applications were studied. The applications were categorized into 6 broad categories: social & communication, entertainment & media, commerce & shopping, travel & transportation, financial services & payment, health & wellness. Below is a graph that depicts the findings.



Entertainment and media apps were found to have the highest percentage of AI reviews with 30.3%, while travel and transportation apps had the lowest percentage of AI reviews with 10%. Out of all the reviews, 25% of them had an AI score of 20% or more. The AI score for the datasets ranged from 0% all the way to 100%, but there is also a chance for false positives that occur because of the AI detector.

Conclusion

In conclusion, our analysis shows that AI generated content is present among Google Play Store app reviews. Our research highlights how different categories of apps have different AI scores, and how some apps have a higher score than others. It is important to note that further

research and analysis must be done in this area as there could be false positives, and more topics to be explored.